PUBLIC SUBMISSION

As of: March 21, 2025
Received: March 15, 2025
Status: [redacted]
Tracking No. m8a-zt3l-qbtn
Comments Due: March 15, 2025
Submission Type: API

Docket: NSF_FRDOC_0001
Recently Posted NSF Rules and Notices.

Comment On: NSF_FRDOC_0001-3479
Request for Information: Development of an Artificial Intelligence Action Plan

Document: NSF_FRDOC_0001-DRAFT-6390
Comment on FR Doc # 2025-02305

Submitter Information

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Organization: OpenMined Foundation

General Comment

See attached file(s)

Attachments

OpenMined

RFI on the Development of an AI Action Plan

March 15, 2025

To: the Networking and Information Technology Research and Development (NITRD) National Coordination Office (NCO), on behalf of the Office of Science and Technology Policy (OSTP)

Thank you for opening up this RFI and for taking the time to review this response. I am writing on behalf of OpenMined, a non-profit foundation of 17,000+ community members with significant experience facilitating *structured access* to proprietary AI systems and their corresponding components. For the past 7 years, we have developed freely available open-source software infrastructure for this purpose and deployed it with various government, industry, and civil society partners.

American AI Leadership is at Risk

For years, the United States of America has held the top position in the global leaderboard for AI innovation, outpacing the world in machine learning model development, frontier AI research, and private investment.[1] The United States' dominance in these key areas indicates a position of strength, but the People's Republic of China (PRC) is working arduously to close this gap and has succeeded in shortening the United States' lead.

In 2015, the PRC announced a national strategic plan and industrial policy - *Made in China 2025* - to turn China into a manufacturing powerhouse, make the country self-reliant, achieve technological leadership, supply chain dominance, and global economic supremacy. An analysis, published in April 2024 in a Hong Kong-based newspaper, concluded that 86% of the objectives China outlined in their plan had been

[1] The Stanford Institute for Human-Centered AI (HAI) created the Global and National AI Vibrancy Rankings tool which provides a comprehensive global ranking of countries for each year from 2017 to 2023, along with detailed metric-specific comparisons to see how each country performs relative to others. *See* https://hai.stanford.edu/ai-index/global-vibrancy-tool.

achieved.[2] A September 2024 report from then-Senator Marco Rubio found that China had reached, or is near reaching, the technological cutting edge in most of the sectors it has targeted.[3] The report goes on to say, “If Xi Jinping were a fund manager, he would have every reason to be pleased with the performance of this portfolio.”

In 2017, China augmented its national industrial plans with the release of its AI Development Plan, a blueprint charting the country's approach to developing AI and sectoral applications. The stated goal of the plan was to make China the world’s primary AI innovation center by 2030. Given China’s previous performance on national plans, as well as its recent rise in AI research output, presence at top-tier AI conferences, AI patent origins, and production of STEM graduates, China is poised to achieve this goal.

A key factor in China’s ability to achieve this goal and the United States' potential to cede its dominance is an overlooked technical property of modern AI systems. While it might seem that the relationship between low-level algorithmic properties and high-level geopolitical consequences would be distant, in this case the implication is profound.

The supply chain for AI capability consists of three critical resources: data, compute, and algorithms - with human talent driving algorithmic development. This triad creates a crucial bottleneck in AI capability within any nation. Today, an AI model’s capability is currently constrained by the amount of these resources that a single organization can acquire. For example, in the United States, GPT-4.5, Claude 3.7, and Gemini 2.0 are all constrained by the amount of these resources that OpenAI, Anthropic, and Google DeepMind, respectively, can acquire. This constraint exists because AI models are currently architected to be *a singular bundled artifact* - trained by a single entity despite drawing resources from many places. This technical property creates a dire trap that pushes democratic nations, like the United States, toward increasingly anti-democratic measures. We can observe this trap through one of the emerging bottlenecks on the above-mentioned resources: data.

Many companies in democratic nations have already exhausted their supply of publicly available training data[4]. Unlike nations that can simply nationalize private data from their citizens and corporations, democratic nations face a fundamental constraint: they cannot match this resource advantage without violating civil liberties. While China can

[2] *See generally* “Ten years on, the relative success of Beijing's Made in China 2025 plan”.
[3] *See* “The World China Made “Made In China 2025” Nine Years Later”.
[4] OpenAI’s former chief scientist, Ilya Sutskever, announced during his test of time award speech at NeurIPS 2024 that, ”We’ve achieved peak data and there’ll be no more”. *See* generally “OpenAI cofounder Ilya Sutskever says the way AI is built is about to change”.

compel Baidu, ByteDance, Alibaba, Tencent, and SenseTime, among other private firms and its 1.4 billion citizens to share data under their Data Security Law[5], the United States must respect the 4th Amendment and other privacy and property laws, ceding leadership in access to this key resource to China.

This bottleneck reveals how the current AI race leads to an inevitable paradox: to maintain dominance in building and deploying the world's most powerful AI models, democratic societies, like the United States, would need to become less democratic. They would need to compel their citizens and businesses to relax individual property rights (e.g. copyright), and to surrender their private data to train competitive models, all in the name of national security and economic competitiveness.

The technical reality here creates an impossible choice for the United States: either lose the AI race or win it by becoming more like what we seek to defeat. A nation whose bedrock values are individual liberty, free markets, and democratic accountability cannot simultaneously build the most capable AI (which requires nationalizing data and compute resources).

This trap is particularly insidious because it emerges not from any single policy choice, but from the inherent nature of AI systems as centralized artifacts requiring centralized control. Understanding that these systems fundamentally aggregate data, compute, and algorithmic capabilities into central models reveals why their development inevitably pushes toward centralization of power. Democratic nations pursuing this path will find themselves inexorably drawn toward increasingly anti-democratic measures, not because anyone desires this outcome, but because the technical architecture itself demands it. This is not merely a moral dilemma - it is a fundamental strategic reality that emerges directly from the technical constraints.

The United States cannot win an AI race that requires us to abandon democracy itself.

Rewrite the Rules of the AI Race

Instead of leaning into their advantages, democratic societies have been competing in a race that amplifies their strategic weakness. This weakness stems not from having fewer of these key AI resources but from bundling them into centralized models. Currently, an American AI model can only be as powerful as the data, compute, and

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[5] While this is a hypothetical scenario, the PRC arguably has these powers under the Data Security Law. *See* https://digichina.stanford.edu/work/translation-data-security-law-of-the-peoples-republic-of-china/

talent that a single company can amass. A Chinese model can only leverage what its government can command. But there exists a third way - one that could harness the distributed resources of the entire free market for information while preserving individual agency and democratic values.

Consider a new distributed infrastructure with an interoperable protocol layer enabling controlled data collaborations across organizations. This infrastructure would preserve data under its original owner's control through source-specific partitioning rather than centralized merging, enabling data owners to selectively participate in specific AI predictions while maintaining governance over their information. Like traditional AI models, such an infrastructure would facilitate rapid knowledge synthesis, but in this new paradigm AI models would combine distributed information only when needed and while preserving source attribution throughout computation. AI models leveraging such a distributed infrastructure for training, instead of a centralized one, would be far more powerful than any single model leveraging resources from a single company or government, because, in such a distributed infrastructure, models can leverage resources from anyone participating in the global network.

Fortunately, this third way is not merely a theoretical possibility. Recent breakthroughs across multiple fields - from cryptography to distributed systems to network theory - offer concrete tools to build such a distributed infrastructure. Advances in privacy-enhancing technologies (PETs) also offer concrete solutions to develop the necessary protocol layer to power the distributed information system. Such an infrastructure would allow the United States to redefine the rules of the AI race and compete in an environment where we make full use of our inherent advantages.

Securing American Leadership in AI

The infrastructure and protocol mentioned above are still in the early stages of research and development, going through proof-of-function pilot projects. The end-to-end ecosystem envisioned does not yet exist, but most of the individual pieces of technology necessary to power such an ecosystem do exist and are ready for further testing.

A central component of the AI Action Plan should prioritize funding the research, infrastructure development, and testing required to support this distributed infrastructure as a core feature of strengthening the United States' national AI capability and maintaining its competitive edge in AI innovation. Such testing should be concretized in technical standards after successful demonstrations of the end-to-end system are complete. Through bold investments like these, the Trump Administration can sustain and enhance America's global AI dominance to promote human flourishing, national security and economic prosperity for the nation for decades to come.

China's desire to be the world's primary AI innovation center by 2030 only works if AI resources cannot be easily shared amongst a group wider than a corporation or nation. History has taught time and time again that free networks beat nations. Case in point, before the internet, the USSR deployed massive resources to catch up with the United States' dominance in mainframe computing. In response, the United States changed the rules of the game and started networking their computers together through a series of pilots across organizations around the country in a program that changed the world: the Advanced Research Projects Agency Network, better known as ARPANET. The DOD established ARPANET, which developed a new technology, which later became the TCP/IP protocol we know and use today. ARPANET's creation was motivated, in part, to allow ARPA-sponsored researchers at various corporate and academic institutions to utilize computers provided by ARPA to communicate with each other and, in part, to distribute new software quickly. In short, the creators wanted to enable resource sharing between remote computers.

Access to ARPANET was expanded through an NSF-funded project called NSFNET which was a program of coordinated, evolving projects to promote research and education networking in the United States. The program created several nationwide computer networks in support of these initiatives, initially aiming to create an academic research network to facilitate access for researchers to NSF-funded supercomputing centers, but through further public funding and private industry partnerships, NSFNET developed into the backbone of the modern Internet.

The technological advancements and practical applications achieved through the ARPANET project under the DOD and the NSFNET project under the NSF resulted in American technological dominance — not by building the biggest computer, but by networking the entire world's computers in a free and democratic web — the Internet.

Today, we can do the same with AI. China can force its 1.4 billion people to share their data with their national AI projects, but they can't force the world's other 6.6 billion people to trust it or use it. By establishing the standard protocol for trusted AI collaboration, America can once again turn its democratic values into technological advantage. Just as the open protocols of the Internet unleashed a wave of innovation that left closed national networks behind, America can build the protocol that will make centralized AI obsolete and carry on the centuries-long trend toward greater freedom and democracy around the world through superior technologies.

The window for action is now. China is betting everything on size. We need to change the game before they realize they're playing the wrong one. Just as ARPANET transformed computing from a race for bigger machines into a democratic network that changed the world, America needs to transform the AI race from a contest of size into a collaboration that amplifies human potential. The future of intelligence is networked, not centralized. Let's build it.

Warmly,

Lacey Strahm
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